Lab 24: Read the catalogue before opening a single file

Topic 2: Tool Design & MCP Integration · Task 2.4: Integrate MCP servers into Claude Code and agent workflows

Runs on Simulator 4: Developer Productivity with Claude · Self-practice

Source: Built from exam guide task statement 2.4 (no practice question covers this angle)

The question

An agent lands in an unfamiliar repo that has a documentation server attached. How should it find out what exists?

A. Expose a resource catalogue so it knows what exists first.

B. The agent greps blindly for the billing runbook.

Simulate the solution

```
# In the browser: open Simulator 4, then Lab 24
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 24

# In a terminal, from simulators/simulator-4-developer-productivity/
uv run explorer.py --case UC-D2
```

What you should see

list_catalog first, read_resource docs://runbook next, no Glob over docs/.

list_catalog precedes read_resource; no Glob or Bash ls touches docs/.

Open `simulators/simulator-4-developer-productivity/evidence/` and find it yourself.

The answer

A. Expose a resource catalogue so it knows what exists first.

Check yourself

- list_catalog is the first tool call
- read_resource docs://runbook follows
- no Glob/Read/Bash over docs/

Now change one thing

Edit the prompt_template in this lab's file under cases/, rerun with --case, and check tool_log.jsonl for what changed.